

$$3) u = x^2$$

$$du = 2x dx$$

$$du = e^{-x} dx$$

$$u = -e^{-x}$$

$$\begin{aligned}\int x^2 e^{-x} dx &= x^2(-e^{-x}) - \int (-e^{-x})(2x) dx \\ &= -x^2 e^{-x} + 2 \int x e^{-x} dx\end{aligned}$$

$$u = x$$

$$du = dx$$

$$du = e^{-x} dx$$

$$u = -e^{-x}$$

$$\int x e^{-x} dx = -x e^{-x} - e^{-x}$$

$$\boxed{\int x^2 e^{-x} dx = -e^{-x} (x^2 + 2x + 2) + C}$$